

9. ETL stands for Extract, Transform, and Load. It is a process used to prepare data for analysis.

Extract: Data is collected from different sources such as databases, spreadsheets or applications.

Transform: The extracted data is cleaned, corrected, formatted and possibly combined with other data.

Load: The transformed data is loaded into a destination such as data warehouse.

ETL is important because analytical results are more useful when the underlying data is accurate, consistent and properly organised. Modern data warehouse architectures commonly use ETL to prepare data for analytics.

10. Common data storage formats

CSV: Stores tabular data using commas to separate values. It is simple and commonly used in spreadsheets and transferring structured data.

JSON (JavaScript Object Notation): Stores data using key-value pairs and nested structures. It is commonly used by APIs and web applications.

Parquet: A column-oriented storage format designed for efficient analytical processing. It is particularly useful for large datasets.

Avro: A row-oriented data serialisation format that stores data together with its schema. It is useful in data-processing systems and data pipelines.

The choice of format depends on the types and size of the data and how the data will be processed.

11. Artificial Intelligence (AI) refers to computer systems designed to perform tasks that normally requires human-like intelligence, such as recognising patterns, understanding language, making predictions and solving problems.

Narrow AI, also called weak AI, is designed to perform specific tasks. Eg. A recommendation system that suggests products to customers.

General AI, also called Strong AI, refers to a theoretical form of AI capable of performing a broad range of intelligent intellectual tasks like at a human-like level.

12. Machine Learning (ML) is a branch of AI in which computers learn patterns from data and use those patterns to make predictions or decisions.

Traditional Programming generally involves a programmer providing explicitly instructions and rules to a computer. Machine learning instead uses data and algorithms to learn patterns.

- Supervised learning! The model learns from labelled data e.g. Predicting house prices using historical house data.

Unsupervised learning! The model identifies patterns in data without predefined labels. e.g. grouping customers according to similar purchasing behaviour.

Reinforcement learning! An agent learns through interaction with an environment and receives rewards or penalties based on its actions.

23. Prompt engineering is the process of designing clear and effective instructions for an AI system in order to obtain more useful and relevant results

It is important because vague prompts can produce vague or incomplete responses, while clear prompts provide AI with sufficient context and requirements

Poor Prompt: "Explain databases."

Improved prompt: "Explain what a database is to a beginner data analytics student and provide two simple real world examples."

24. The MAP function is a programming function used to apply another function to each item in a collection of data.

E.g., in Python:

```
<1> Python
```

```
Numbers = [1, 2, 3, 4]
```

```
Squared = list(map(lambda x: x * x, numbers))
```

The map function applies the operation to every number, producing [1, 4, 9, 16]

This can be useful in data processing when the same operation needs to be applied to multiple values.

25. Cloud computing is the delivery of computing resources such as storage, servers, databases and software over internet.

IaaS - Provides infrastructure such as virtual machines, storage and networks. e.g. Amazon EC2.

PaaS - Provides a platform on which developers can build and deploy applications without managing all the underlying infrastructure. e.g. Google App Engine.

SaaS - Provides complete software application through the ~~intern~~ internet. e.g. Microsoft 365.

26. A version control system (VCS) is a system that records changes made to files overtime. It allows users to track previous versions, collaborate and restore earlier versions when necessary.

Git is useful to data analysts and developers because it allows them to track changes to code, notebooks and project files and collaborate through platforms like Github.

A commit is a recorded snapshot of changes in a repository.

A branch is a separate line of development that allows users to work on changes without immediately affecting the main development line.

27. A Jupyter notebook is an interaction environment that allows users to combine programming code, written explanations, data visualisation and results in one document.

It is mostly associated with ~~pyto~~python, although jupyter supports other programming languages.

Jupyter notebooks are popular among data analysts and scientists because they allow users to write and run code step-by-step while immediately viewing results, charts, and analysis.

28. Python is a high level programming language known for its readable syntax and wide range of application.

Pandas : Used for data manipulation and analysis

Numpy : Used for numerical calculation and working with ~~any~~ arrays.

Matplotlib : used to create data visualisation like charts / graphs.

Scikit : Used for machine learning tasks like regression and clustering.

29. Data Privacy refers to the appropriate handling and data protection of personal information.

Data privacy is important because individuals should have control over how their personal information is used and organisations have responsibilities when handling that responsibility.

The General Data Protection Regulation is a European Union regulation concerning the protection of personal data and privacy.

30. Data Ethics refers to principles and standards that guide responsible^h collection, use and analysis of data.

1. Collecting personal information without concern
2. Using biased data to make ~~informed~~ important decision
3. Selling or sharing personal information without authorisation

31. Data Governance is the framework of policies, processes, roles and responsibilities used to manage data effectively throughout an organisation.

Data quality: Ensuring data is accurate, complete and consistent.

Data security: Protecting data from unauthorised access.

Data privacy: Ensuring personal information is handled appropriately.

Data ownership: Establishing consistent responsibilities for different datasets.

~~Data~~

32. Bias in AI occurs when an AI system produces systematically unfair outcomes for certain individuals or groups.

Potential consequences include unfair identification, discrimination and incorrect decisions.

AI bias comes from factors like biased training data, incomplete datasets or the way algorithms are designed.

33. Data Role	Responsibility	Skills
Data Analysts	Analyses data to produce insights and report	SQL, Excel, statistics, Visualisation
Data scientist	Build models and perform advance analysis	statistics, ML, Python, mathematics.
Data engineer	Builds and maintain data infrastructure and pipelines	SQL, Python, databases, cloud.
AI engineer.	Develops and deploys AI powered systems	Python, ML, AI, software engineering.

34. Business intelligent refers to technologies and processes used to analyse organisational data and present information that supports business decision making.

Power BI - used to connect data sources, analyse information and create interactive reports and dashboards.

Tableau - used to explore data and create interactive visualisations.

35. The concept I found most interesting was the difference between a data warehouse and a data lake. Before studying these words, I mainly thought of data storage as simply keeping information in a database.

36. The concept I found most difficult was ETL because it involves several stages of working with data. I initially found it difficult to understand why data needs to be transformed before loaded into another system.

37. I believe that data and AI will have a significant impact on my future career because organisations increasingly rely on data to make decisions.

AI is also likely to influence the workplace by automating repetitive tasks and assisting people with analysing information. Learning how to work alongside AI while developing strong data analytics skills will help me remain adaptable as technology continues to develop.

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21. REST stands for Representational State Transfer. It is an API architecture that commonly uses HTTP methods to interact with resources.

GET: Retrieves data

POST: Creates or submits new data

PUT: Updates or replaces existing data

DELETE: Removes data.

Eg. A data application could use GET to retrieve customer information and POST to add a new customer.

22. JSON is a lightweight format used to represent and exchange structured data.

JSON is commonly used with APIs because it is relatively easy for humans to read and for computer systems to process.

Fig. <1> JSON

```
{  
  "name": "Thabone",  
  "age": 20,  
  "occupation": "Data Analyst"  
}
```

19. An API is a mechanism that allows different softwares applications or systems to communicate and exchange information.

A simple analogy is a waiter in a restaurant. The customer a request to the waiter, the waiter communicate the request to the kitchen, and the results is return to the customer.

Simple Similarly, an API receives a request from one application, communicates with another system and returns the appropriate response.

APIs are important because they allow different applications and services to work together without requiring one system to know all the internal details of another system.

20. An API key is a unique identifier or credential used by an API provider to identify and control access to its services.

API keys should be treated as sensitive information. They should not be publicly posted on websites, Github repositories or shared unnecessarily. They should instead be stored securely using appropriate methods such as environment variables or secret-management system.

If an API key is exposed, another person could potentially use the associated service under the owner's account, potentially resulting in unauthorised access or financial costs.

17. A training dataset is a collection of data used to teach a machine learning or AI model to recognise patterns and perform a particular task.

The quality of training data is important because an AI system learns from the information provided to it. If the data is incomplete, inaccurate or biased, the resulting model may produce unreliable or unfair results.

Example

If an AI recruitment system is trained using historical data containing discriminatory patterns, the system may reproduce ~~those~~ those patterns when evaluating candidates.

18. Computer Vision is an area of AI that enables computers to analyse and interpret information contained in images and videos.

A Computer-vision system can process visual information and identify objects, patterns, shapes or other features.

Examples

1. Facial recognition: Identifies or verifies people based on facial features.

2. Medical imaging: Helps analyse medical images to identify patterns that may assist healthcare professionals.

15. A Large Language Model is an AI Model trained on large quantities of text to understand and generate human-like language.

Unlike a standard machine learning model designed for a specific prediction or classification task, an LLM can perform a wide range of language-related tasks such as answering questions, summarising information, translating text and generate content.

Example

GPT: Used for conversation, reasoning, writing and other language tasks.

Claude: Used for tasks such as writing, analysis, summarisation and question answering.

16. Generative AI refers to AI systems capable of creating new content based on patterns learned from training data. Depending on the system, this can include text, images, videos or computer code.

At a high level, a generative AI learns patterns from large amount of training data and uses those patterns to generate new outcomes in response to an input or prompt.

Examples

1. Chat GPT: Generate and analyses text.
2. Gemini: Assists with text, reasoning and other multimodal tasks.
3. Microsoft Copilot: Uses AI to assist users with tasks such as writing, information retrieval and productivity.

13. Deep learning is a specialised area of machine learning that uses multi-layered neural networks to learn complex patterns from large amounts of data.

Related like this: Artificial Intelligence - machine learning - Deep learning -

A neural network is a computational model inspired by the way biological brains process information. It contains interconnected nodes arranged into layers. Data passes through these layers, and the network adjusts its internal parameters during training to improve predictions.

14. Natural Language Processing is a field of AI that enables computers to process, understand and generate human language.

1. Chatbots: Understand and respond to written questions.
2. Voice Assistance: Convert spoken language into commands and responses.
3. Email spam filters: Analyse language patterns to identify unwanted messages.

7. Data mining is the process of examining large datasets to discover useful patterns, relationship or trends.

Businesses use data mining to support decision making, identify customer behaviour, detect unusual activities and improve their operations.

Examples

1. A supermarket can analyse purchasing patterns to determine which products customers frequently purchase together.
2. A bank can analyse transaction pattern to identify potentially suspicious or unusual transactions.

8. A Database Management System is a software that allows users and applications to create, store, organise, retrieve and manage data in databases.

DBMS	Description	Suitable use
MySQL	open source relational DBMS	Websites and business application
PostgreSQL	Advanced open source relational DBMS	Complex application and analytical workload
Oracle database	Enterprise level relational DBMS	Large organisational & critical business system.
MongoDB	NoSQL document oriented database	Applications handling flexibility or semi-structured data.

5. Structured data is data organised according to a ~~predet~~ ~~preferred~~ ~~format~~ predefined format, usually rows and columns.

E.g. A customer table containing customer names, account numbers and balances.

Semi-structured data does not follow a traditional table structure but contains organisational elements such as tags, keys or metadata. E.g. JSON file containing customer information.

Unstructured data does not follow a predefined data structure. E.g. Images, videos, audio recordings and free-form documents.

6. Data modelling is the process of designing how data will be organised, related and stored within a system. It is important because a good data model helps ensure data is organised logically, reducing unnecessary duplication and makes data easier to access and analyse.

- Relational data model organises data into tables containing rows and columns. Relationships can be established between tables using keys.

- Dimensional data model organises data into ~~fact tables~~ structures such as fact tables and dimension tables. It is commonly used in data warehouses and business intelligence because it supports analytical queries.

3. A data lake is a centralised storage repository that can store large amounts of data in its original or raw format. It can contain structured, semi-structured and unstructured data, such as tables, JSON files, images, audio and video.

A data warehouse normally requires data to be structured before or as it is loaded, whereas a data lake can retain data in its original form.

A business may prefer a data lake when it has a very large amounts of different types of data and wants flexibility for future analytics, machine learning or data science.

4. A data mart is a smaller, focused collection of data designed to serve a particular department or business function.

A data mart can be created from a large data warehouse, e.g. a company may have a data warehouse containing information from the entire organisation, while its finance department has a finance data mart containing only financial information.

Data marts are typically used by specific departments such as finance, marketing, sales, or human resources.

Brightleorn Data Analytics Research Assignment

1. A database is an organised collection of data that allows information to be stored, managed, retrieved and updated efficiently. Databases are essential in the modern world because organisations generate and manage large amount of information everyday. A database makes it easier to access accurate information and supports business operations and decision making.

Example

1. Banks use databases to store customer details, account information and transaction records.
2. Hospitals use databases to store patient information, medical records and appointment details.

2. A data warehouse is a centralised system used to collect and store data from multiple sources, mainly for reporting, analysis and decision making. Unlike a regular operation database, which is primarily designed to handle day-to-day transactions, a data warehouse is designed to analyse large amount of historical data.

A company may use a data warehouse to combine information from sales, finance, marketing and customer system. Managers can then analyse this information to identify trends and make informed decisions.

Database

- Mainly support daily operations.
- Frequently updated
- Handles transactions

Data Warehouse

- Mainly supports analysis
- Stores large amount of historical data.
- Handles analytical queries